Project Report

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Product Recommender System

Course: Data Warehousing and Data Mining (AD211)
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Table of Contents

1. **Introduction**
2. **Problem Statement**
3. **Motivation**
4. **Aims and Objectives**
5. **Methodology**
6. **Results and Discussion**
7. **Conclusion**
8. **References**

1. Introduction

Recommender systems have become an essential part of e-commerce, media platforms, and many other industries, helping users find relevant items based on their preferences. This project implements a recommender system using different machine learning algorithms: **K-Nearest Neighbors (KNN)**, **Singular Value Decomposition (SVD)**, and **Random Forest Regression**, evaluated over a product-rating dataset.

2. Problem Statement

To develop a recommendation system that can predict a user's interest in a product by analyzing past user-product interaction data. The objective is to:

- Filter users and products with insufficient data.
- Compare various models.
- Recommend top-N products for a given user.

3. Motivation

As users are overwhelmed by choices online, recommendation systems help in reducing decision fatigue by suggesting relevant items. Understanding and building such systems provides hands-on insight into collaborative filtering, matrix factorization, and regression techniques.

4. Aims and Objectives

Aim:

To develop and evaluate a recommender system using machine learning algorithms that can predict user preferences and provide personalized product recommendations based on historical rating data.

Objectives:

1. To explore and implement multiple recommendation strategies, including:
    - Collaborative Filtering using K-Nearest Neighbors (KNN)
    - Matrix Factorization using Singular Value Decomposition (SVD)
    - Supervised learning using Random Forest Regressor
2. To design a data preprocessing pipeline that filters sparse and noisy user-product interactions.
3. To utilize a relational database (SQLite) for efficient storage and querying of user-rating data.
4. To compare the accuracy and efficiency of the implemented models using relevant performance metrics and visualizations.

5. To simulate OLAP-like filtering operations to enhance data quality before model training.
6. To generate top-N personalized product recommendations for a given user.
7. To interpret and visualize model outputs using plots and heatmaps for better understanding.

5. Methodology

The methodology followed in this project encompasses multiple stages, from data ingestion and cleaning to model training and evaluation. Each stage was carefully designed to ensure a robust recommendation system using different machine learning techniques. The key steps are outlined below:

1. Data Collection and Storage

- A dataset containing user ratings for products was used as the basis for recommendation.
- The dataset was imported from a CSV file and stored in an SQLite relational database for ease of querying and efficient data handling.

2. Data Preprocessing

- Non-numeric and missing rating values were cleaned and converted appropriately.
- Users and products with fewer than a minimum threshold of reviews (e.g., 3) were filtered out to reduce noise and improve model performance.
- A user-item rating matrix (pivot table) was constructed with users as rows, products as columns, and rating values as entries.

3. Exploratory Data Analysis (OLAP-style)

- Summary statistics and distribution plots were generated to understand user behavior and rating tendencies.
- Visualizations such as histograms and heatmaps were used to analyze review density and patterns in the pivot matrix.

4. Model Implementation

Three recommendation approaches were developed and tested:

- K-Nearest Neighbors (KNN):
  Collaborative filtering based on item-item similarity using cosine distance.
- Singular Value Decomposition (SVD):
  Dimensionality reduction was performed to extract latent features. These were used to reconstruct the rating matrix and generate recommendations.

- Random Forest Regressor:
  A supervised model was trained on user rating vectors to predict average user ratings, which were then used to recommend top-rated items.

5. Model Training and Prediction

- Each model was trained using the preprocessed pivot matrix.
- SVD and Random Forest outputs were inverse transformed or reshaped into complete prediction matrices.
- Recommendations were generated for specific products based on similarity scores or predicted user preferences.

6. Visualization and Evaluation

- Model performance was evaluated using metrics such as RMSE (Root Mean Square Error) and R^2 score.
- Plots of cumulative explained variance (SVD), rating distributions, and pivot heatmaps were generated to illustrate model behavior and data characteristics.

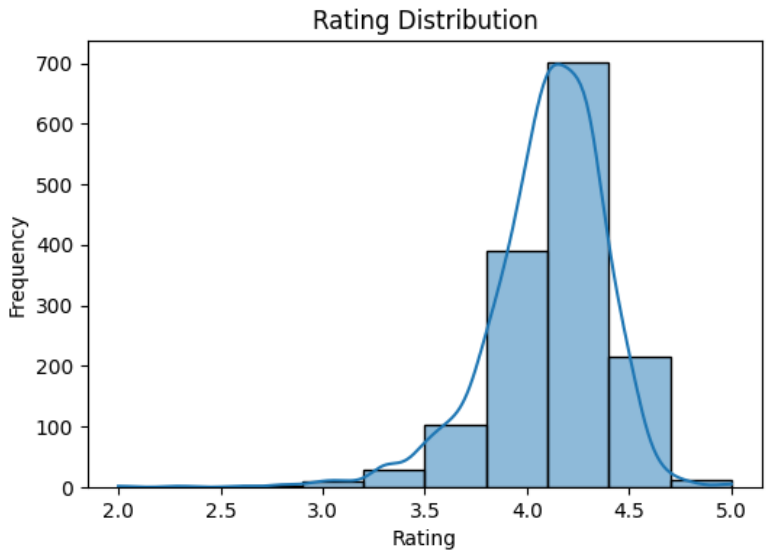
7.Tools Used

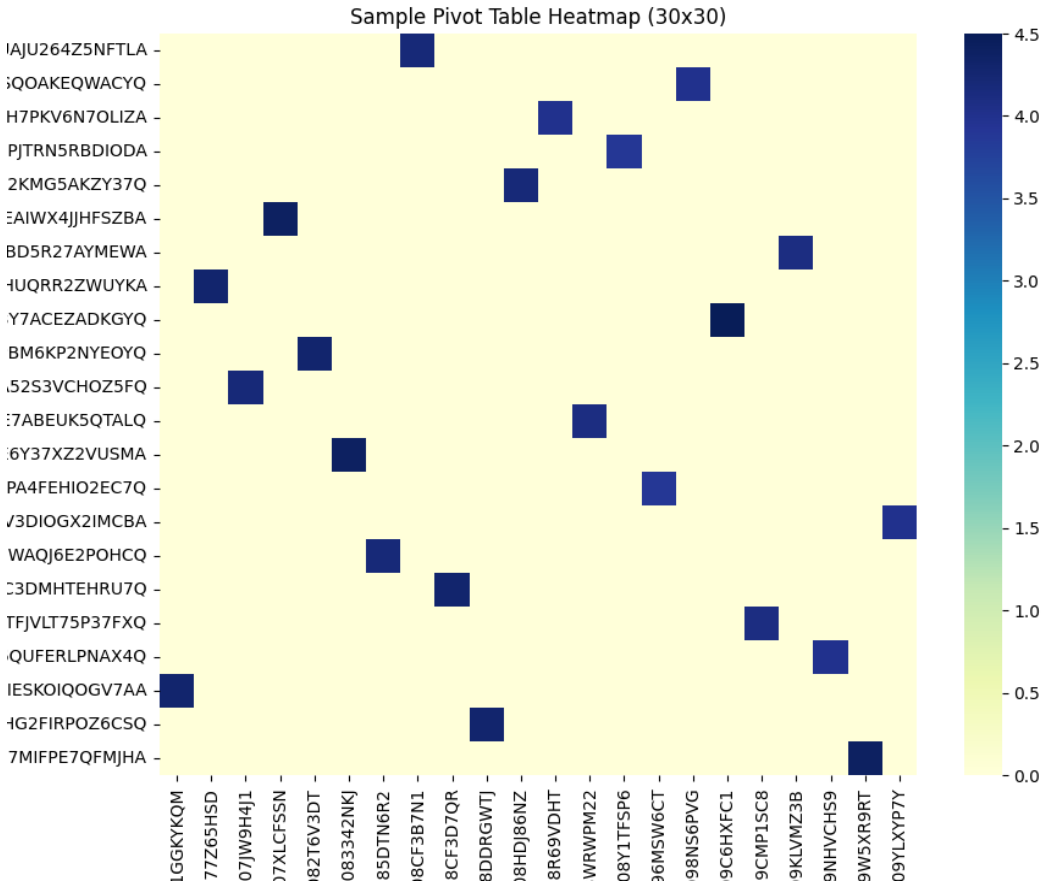
Python, Pandas, NumPy, Scikit-learn, Surprise, SQLite3, Matplotlib, Seaborn

6. Results and Discussion

6.1 Data Distribution

- **Rating Histogram**: Distribution of rating values across the dataset.
- **Heatmap**: Shows sparsity of the rating matrix.





6.2 Model Outputs

- **SVD Explained Variance Plot**
- **KNN Similarity Matrix (heatmap)**
- Sample output of recommended products for a user (e.g., U100):

```
PS C:\Users\charu\Desktop\DWDMProject> python -u "c:\Users\charu\Desktop\DWDMProject\finalcheck.py"
2025-05-05 22:10:25,819 INFO:Imported CSV into SQLite table 'amazon_reviews' (1464 rows)
2025-05-05 22:10:26,252 INFO:Loaded 1464 rows from SQL table 'amazon_reviews'
2025-05-05 22:10:27,034 INFO:Pivot matrix shape: (22, 22)
2025-05-05 22:10:27,736 INFO:Trained KNN model.
2025-05-05 22:10:27,870 INFO:Trained Random Forest model.
2025-05-05 22:10:28,050 INFO:Trained SVD model and plotted explained variance.
Top 10 products by average rating: ['B09C6HXFC1', 'B083342NKJ', 'B09W5XR9RT', 'B07XLCFSSN', 'B077Z65HSD', 'B01GGKYKQM', 'B08DDRGWTJ', 'B0
82T6V3DT', 'B08CF3D7QR', 'B07JW9H4J1']
Enter a product ID for recommendations: B096MSW6CT
KNN Recommendations: ['B01GGKYKQM', 'B07JW9H4J1', 'B077Z65HSD', 'B082T6V3DT', 'B083342NKJ', 'B085DTN6R2', 'B07XLCFSSN', 'B08CF3B7N1', 'B0
8CF3D7QR', 'B08HDJ86NZ']
SVD Recommendations: ['B09YLXYP7Y', 'B077Z65HSD', 'B09NHVCHS9', 'B08WRWPM22', 'B07JW9H4J1', 'B08HDJ86NZ', 'B08CF3D7QR', 'B08DDRGWTJ', 'B0
82T6V3DT', 'B085DTN6R2']
Random Forest Recommendations: ['B08CF3D7QR', 'B083342NKJ', 'B08WRWPM22', 'B09YLXYP7Y', 'B08CF3B7N1', 'B08DDRGWTJ', 'B09NHVCHS9', 'B09W5X
R9RT', 'B09C6HXFC1', 'B08HDJ86NZ']
Evaluation metrics: {'RF_MSE': 1.700537190082719e-05, 'RF_RMSE': np.float64(0.004123757012825463), 'RF_R2': 0.8058820754716898}
PS C:\Users\charu\Desktop\DWDMProject>
```

7. Conclusion

The project effectively demonstrates a comprehensive recommendation pipeline leveraging both filtering techniques and machine learning models on real-world data. while KNN offered an intuitive approach through user similarity, albeit with some limitations. Random Forest added flexibility with its ensemble-based regression capabilities. Overall, the models highlight complementary strengths, and future work could focus on developing hybrid recommendation systems and deploying the solution as a web application.

8. References

Amazon Ratings

- Source: Kaggle – Amazon ratings

Python Libraries and Tools Used

- Python Software Foundation. *Python Language Reference, version 3.x*. Available at: https://www.python.org.